

Scalability & Future Plan

Date	3 July 2026
Team ID	SWTID-2026-9068
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document explains how the **Smart Lender** system can be enhanced to support more users, larger datasets, improved prediction accuracy, and advanced banking features. It also outlines future enhancements to make the application more secure, scalable, and suitable for real-world financial institutions.

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local server.	Deploy the application on cloud platforms such as AWS, Azure, or Google Cloud with load balancing and auto-scaling.
2	Data Storage	Stores user details and prediction records in a local database.	Use scalable cloud databases such as PostgreSQL or MongoDB with automated backup and recovery.
3	Performance	Loan predictions are generated in real time for each request.	Implement caching, optimized ML models, and asynchronous request processing to improve response time.
4	Security	Basic user authentication and secure data storage.	Implement OAuth authentication, data encryption, role-based access control, and secure REST APIs.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited support for concurrent users	System performance may decrease when many users submit loan applications simultaneously.	High
2	Limited integration with banking systems	Users cannot directly verify bank accounts or access real-time financial information.	High
3	Prediction accuracy depends on available training data	Insufficient or outdated datasets may reduce the accuracy of loan eligibility predictions.	Medium

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integration with banking APIs, Aadhaar/PAN verification, and credit score services.	2 Months	Enables faster customer verification and more accurate loan eligibility assessment.
Phase 3	AI-powered loan recommendation, EMI calculator, fraud detection, and document verification.	4 Months	Improves decision-making, reduces fraudulent applications, and enhances user experience.
Phase 4	Mobile application, multilingual support, cloud deployment, analytics dashboard, and chatbot assistance.	6 Months	Increases accessibility, improves customer support, and enables large-scale deployment for banks and financial institutions